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EE1060: Differential Equations and Transform Techniques

Lecture 23: Introduction to Fourier Transform

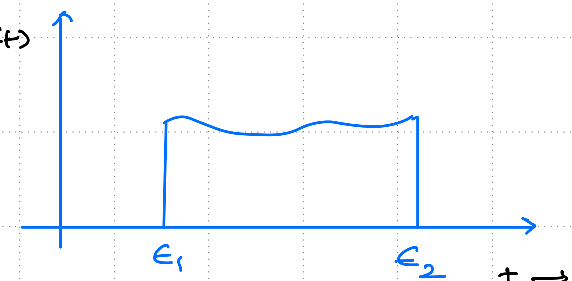
Topics :

1. From Fourier Series to Fourier Transform
2. Inverse Fourier Transform

Fourier Series of Non-Periodic Signals - Idea of Periodic Repetition

Fourier Series $\longrightarrow$ Synthesize a periodic signal in terms of complex exponentials

Non-periodic signals $\longrightarrow$?



$x(t) \longrightarrow$ original signal

$y(t) \longrightarrow$ signal with periodic Repetition.

Periodic Repetition?

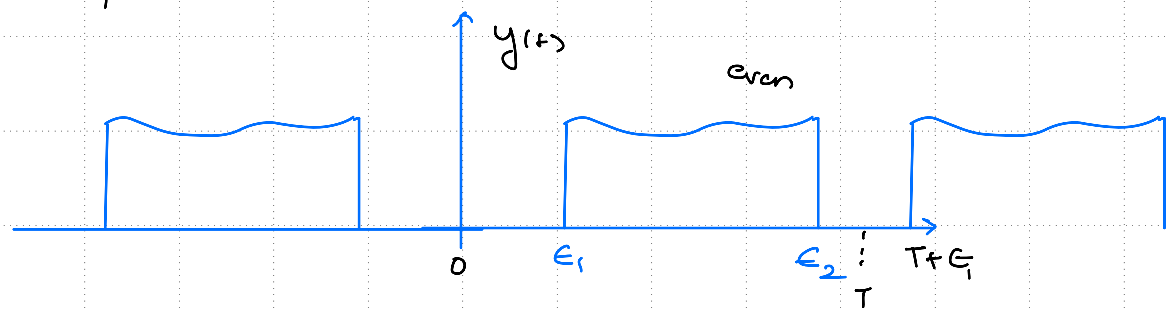
$(T) \leftarrow$ arbitrary

① Repeat such that

Cos terms $\leftarrow$ $y(t)$ is even (or)

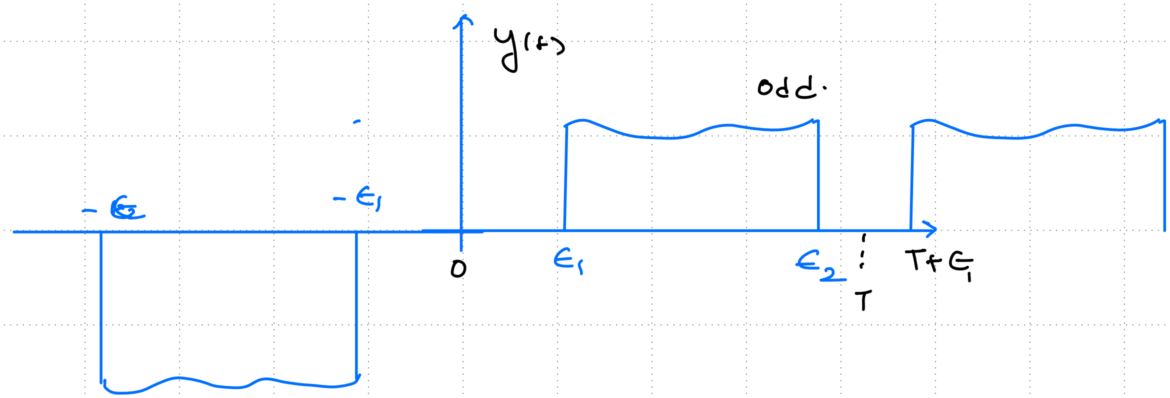
Sin terms $\leftarrow$ $y(t)$ is odd

may not be significant



② Let $T \longrightarrow \infty$

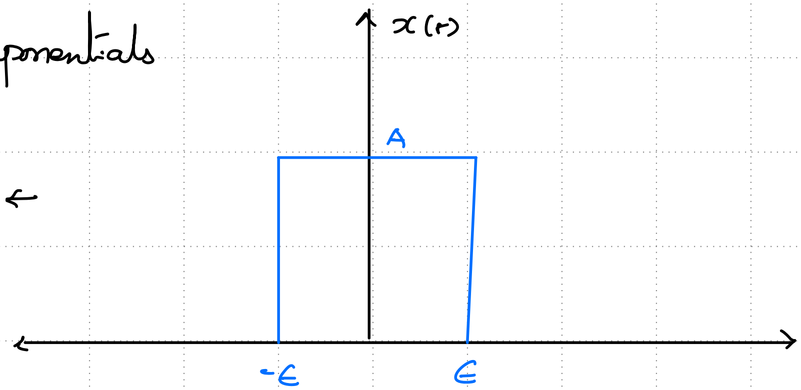
$\underset{(T \to \infty)}{y(t)} \longrightarrow x(t)$



The effect of Varying Time Period

$y(t)$ can be synthesized using complex exponentials

$$y(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T}$$

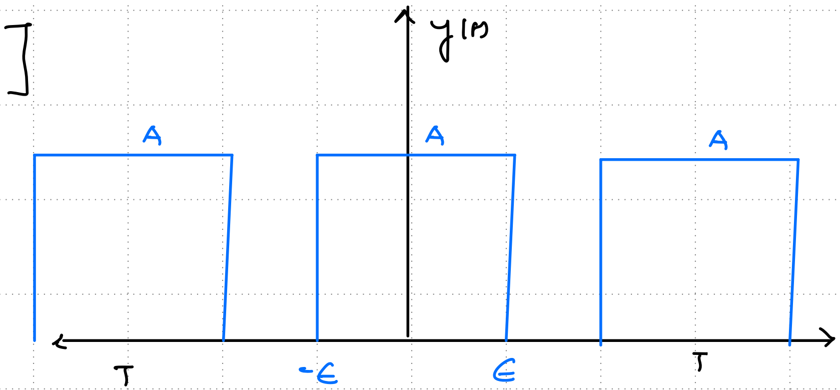


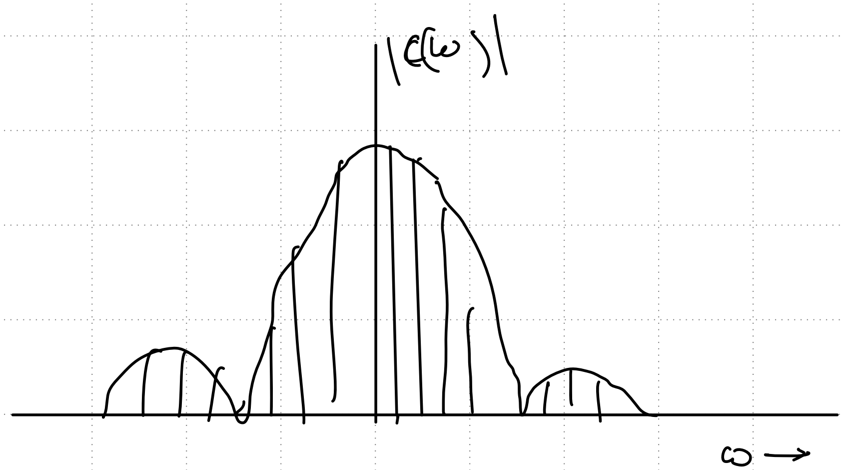
$$C_k = \frac{1}{T}\int_{-\epsilon}^{\epsilon} A e^{-jk\omega_0 t}\,dt \qquad (2\epsilon < T)$$

$$= \frac{A}{T}\frac{1}{-jk\omega_0}\left[e^{-jk\omega_0\epsilon} - e^{jk\omega_0\epsilon}\right]$$

$$= \frac{2A}{Tk\omega_0}\sin(k\omega_0\epsilon) =$$

$$= \frac{2A}{T}\frac{\sin(k\omega_0\epsilon)}{k\omega_0}.$$





$$C_0 = \frac{2\epsilon}{T}A$$

$$C(\omega) = \frac{2A}{T}\frac{\sin(\omega\epsilon)}{\omega} \quad \text{when} \quad \omega = k\omega_0$$

Definition of Fourier Transform and Inverse Fourier Transform

as $T \to \infty$, a) $T\omega_0 \to \downarrow$, gap b/w the bins $\to$ reduce.

b) mag of c_k also changes.

Can we look $T c_k \to ?$

$$X(\omega) = \lim_{T\to\infty} T c_k$$

$$= \lim_{T\to\infty} T \cdot \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, e^{-j \underbrace{k\omega_0}_{\to \omega} t}\, dt$$

$$= \lim_{T\to\infty} \int_{-T/2}^{T/2} x(t)\, e^{-j\omega t}\, dt$$

Fourier Transform of $x(t)$ $\longrightarrow$ $X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

Let $t \to \infty$ $\quad x(t) = \{ 1 \;\; \forall t \longrightarrow X(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega t}\, dt \;?$

$\downarrow$ function of ω

Ans: $X(0) = \text{finite}$

$X(\neq 0) = 0$.

$x(t) \qquad X(j\omega)$

$1 \Longleftrightarrow \delta(\omega) \longleftarrow$

Fourier Transform and Inverse Fourier Transform

$$y(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T}$$

from the Fourier Transform $X(j\omega)$ or $X(\omega)$:-

a) Sample at appropriate points $\rightarrow k\omega_0 \qquad \omega_0 \rightarrow \frac{2\pi}{T}$

$\Rightarrow$ This will give the Fourier Series Coefficients of $y(t)$.

$$= \sum_{k=-\infty}^{\infty} \frac{X(j\omega)}{T} e^{jk\omega_0 t}$$

$$= \sum_{k=-\infty}^{\infty} \frac{X(j\omega)}{2\pi} \omega_0 e^{jk\omega_0 t}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(j\omega) e^{jk\omega_0 t} \omega_0$$

b) $T \rightarrow \infty$, $x(t)$ [Note that in the limiting case $\omega_0 \rightarrow$ much smaller $\hookrightarrow d\omega$ — Crucial.

and summation $\longrightarrow \int$.

$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

In literature FT of $x(t)$ is rep as. $X(j\omega)$ or $X(\omega)$ $\downarrow$ for this Course.